

CHAPTER 03

Force, Mass & Motion

Why does a skateboard roll downhill? Why does a soccer ball curve when you kick it just right? Why do you slide forward when the school bus slams its brakes? Time to meet the rules that govern every moving thing in the universe.



Georgia Standard S8P3

WHAT YOU'LL LEARN

- | | |
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| 01 Motion & Reference Points | 06 Forces & Net Force |
| 02 Distance vs. Displacement | 07 Newton's First Law (Inertia) |
| 03 Speed | 08 Newton's Second Law ($F = ma$) |
| 04 Velocity | 09 Newton's Third Law (Action/Reaction) |
| 05 Acceleration | |

3.1 Motion & Reference Points

Right now, you might think you're sitting still. But are you? Earth is spinning at about **1,000 mph**. It's orbiting the Sun at **67,000 mph**. The whole solar system is racing through the Milky Way at **500,000 mph**. So... are you moving or not?

The answer depends on what you compare yourself to. That's why scientists define motion using a **reference point** — a fixed object or place you compare your position to. **Motion** is a **change in position over time** relative to a reference point. If you compare yourself to your desk, you're not moving. If you compare yourself to the Sun, you're flying through space. Both are technically true. That's the trick of motion — it's **relative**.

! Did You Know?

When you watch a movie at the theater, you and your popcorn are moving through space at over **1.3 million miles per hour** relative to the cosmic microwave background. You'd never know it, because everything around you is moving with you. Motion only feels real when you can see something change.

★ Real-World Connection

Ever sat in a stopped car at a red light, then suddenly thought your car was rolling backward — only to realize the car next to you was moving forward? Your brain used the wrong reference point! Choosing the right reference point matters.

3.2 Distance vs. Displacement

Both **distance** and **displacement** describe how an object moves — but they mean different things.

- **Distance** is the **total length** of the path traveled. It doesn't care about direction. Walked in circles? Add up all those steps.
- **Displacement** is the **change in position** from start to finish — measured in a straight line, with a direction.

Walking around a track: Distance vs. Displacement

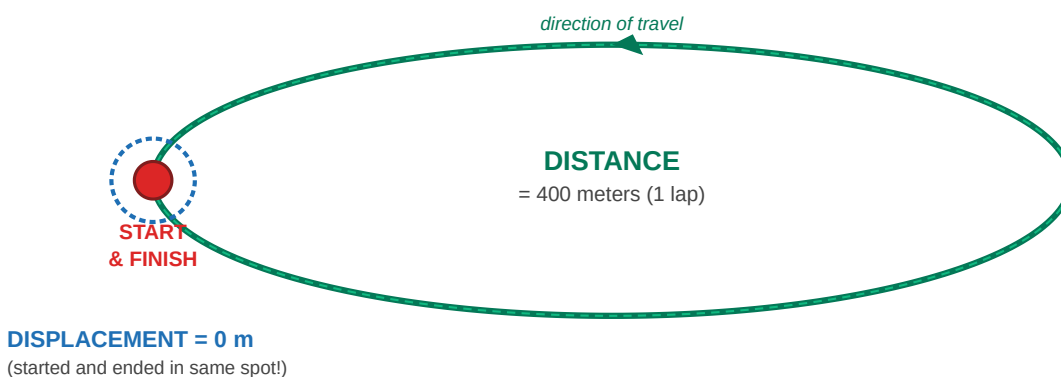


Figure 3.1 — Walk one full lap around a 400-meter track and your distance is 400 m, but your displacement is zero because you ended up exactly where you started.

✗ Common Mix-Up

Distance and displacement *can* be the same number — but only when you move in a perfectly straight line in one direction. The moment you turn or backtrack, the two numbers split. Always ask yourself: "Where did they end up?" for displacement, "How far did they travel total?" for distance.

3.3 Speed

Speed tells us **how fast** something is moving — how much distance it covers in a certain amount of time. The formula is short and sweet:

$$\text{Speed} = \text{Distance} \div \text{Time}$$

measured in meters/second (m/s), miles/hour (mph), etc.

If your car covers 60 miles in 1 hour, your speed is 60 mph. If a sprinter runs 100 meters in 10 seconds, their speed is 10 m/s.

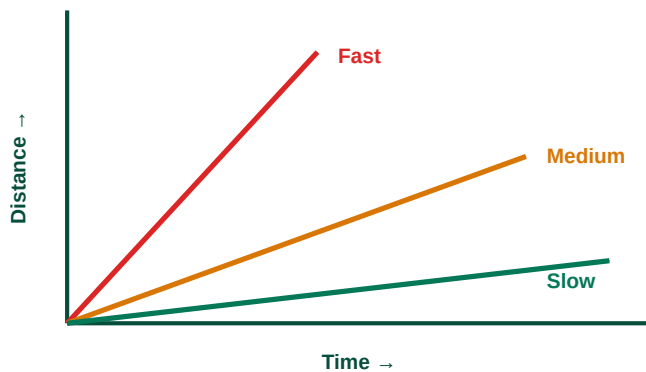
Average Speed vs. Instantaneous Speed

When you're driving, your speedometer shows your *instantaneous* speed — your speed at this exact moment. **Average speed**, on the other hand, is the total distance you traveled divided by the total time it took, even if you sped up and slowed down along the way.

Driving from Woodstock to Atlanta might be 35 miles. If it takes you 1 hour, your average speed is 35 mph — even though you went 70 mph on the highway and 0 mph stuck in traffic.

Reading a Distance-Time Graph

Reading a Distance-Time Graph



Key Rules

Steep slope = **FAST** speed

Gentle slope = **SLOW** speed

Flat (horizontal) = **NOT MOVING**

Straight line = **CONSTANT** speed

Figure 3.2 — On a distance-time graph, the slope *IS* the speed. Steeper = faster.

? Quick Check 3.3

1. A runner travels 200 meters in 25 seconds. What is their average speed?
2. On a distance-time graph, what does a horizontal (flat) line mean about the object's motion?
3. Sketch how a distance-time graph would look for a car that sits still for 5 minutes, then drives at a constant speed for 10 minutes.

3.4 Velocity

Here's a subtle but important distinction. **Velocity** is just speed **with a direction**. That's the only difference.

- "60 mph" = **speed**.
- "60 mph north on I-75" = **velocity**.

It seems like a small change, but direction matters a lot in physics. Two cars moving 60 mph in opposite directions have the same *speed* but very different *velocities* — and if they collide head-on, you'd definitely notice the difference!

! Did You Know?

A car going around a circular track at a constant 50 mph has a **constant speed** but its **velocity is constantly changing** — because its direction keeps changing. Believe it or not, this means the car is technically *accelerating* the entire time, even though the speedometer never moves. You'll see why next.

3.5 Acceleration

Acceleration is the rate at which **velocity changes** over time. In everyday English, "accelerating" usually means speeding up. In physics, it means **any change in velocity** — which means:

- **Speeding up** is acceleration.
- **Slowing down** is acceleration (sometimes called **deceleration** or negative acceleration).
- **Changing direction** — even at constant speed — is acceleration.

$$\text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Time}}$$

measured in m/s^2 (meters per second, per second)

That weird unit " m/s^2 " might look strange, but it just means "how much your speed changes every second." If a sports car accelerates at 5 m/s^2 , it means its speed increases by 5 m/s every single second. After 1 second, it's going 5 m/s . After 2 seconds, 10 m/s . After 3 seconds, 15 m/s . And so on.

Velocity-Time Graphs

Three Types of Motion on a Velocity-Time Graph

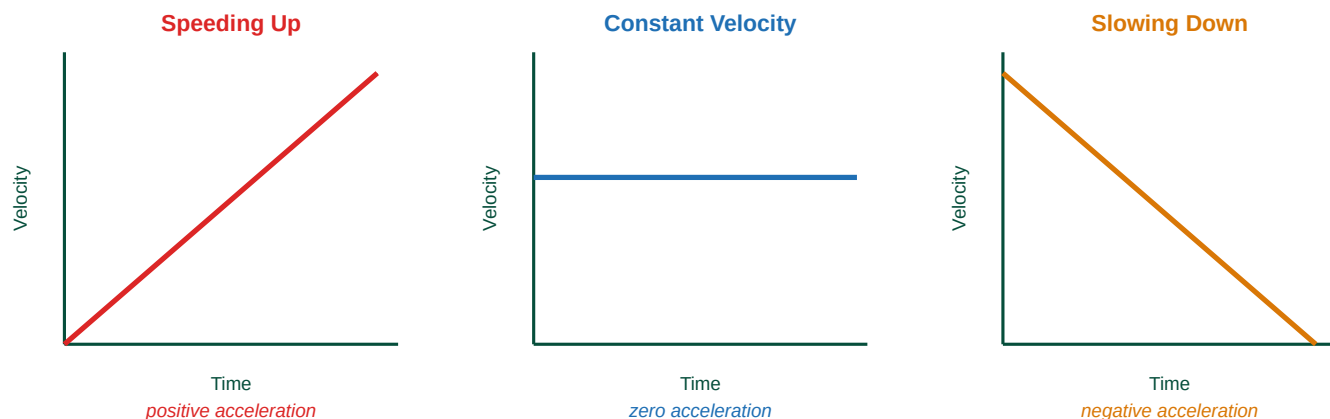


Figure 3.3 — On a velocity-time graph, the slope tells you the acceleration. Up-sloping = speeding up. Flat = constant speed. Down-sloping = slowing down.

★ Real-World Connection

When you ride a roller coaster, you feel that "push back into the seat" sensation as the train speeds up at the start. That's not just speed — it's **acceleration**. Your body feels acceleration much more strongly than it feels constant motion. Once the coaster levels off at top speed, you can barely tell you're moving at all.

? Quick Check 3.5

1. A car goes from 0 m/s to 30 m/s in 10 seconds. What is its acceleration?
2. A car driving 30 mph in a perfectly straight line for 5 minutes — is it accelerating? Why or why not?
3. How does a velocity-time graph show whether an object is speeding up vs. slowing down?

3.6 Forces & Net Force

What actually causes motion to start, stop, or change? **Forces**. A force is simply a **push or a pull**. Every time motion changes — every time something starts, stops, speeds up, slows down, or turns — there's a force doing it.

Forces are measured in **newtons** (N), named after Sir Isaac Newton. One newton is roughly the force needed to hold up a small apple against gravity. (Yes, the same scientist who supposedly got hit by a falling apple has the unit of force named after him.)

Net Force

Usually more than one force acts on an object at the same time. When you push a desk, friction pushes back. When you stand on the floor, gravity pulls you down while the floor pushes you up. The **net force** is the **overall force** when you combine all of them.

Calculating Net Force

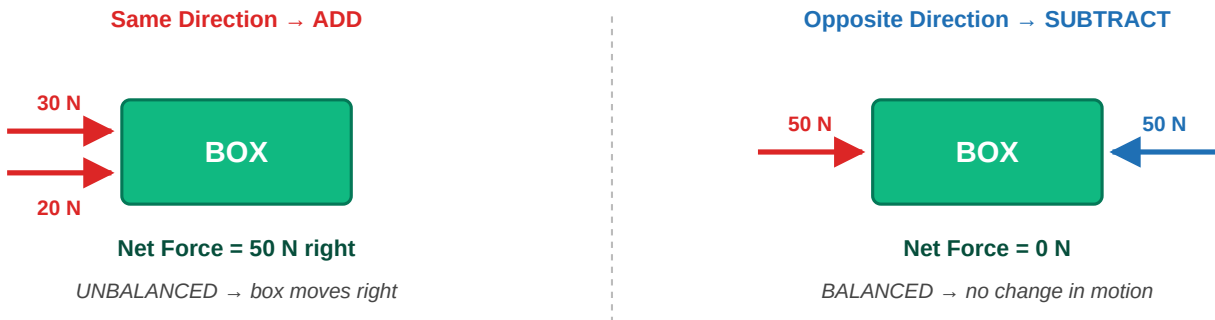


Figure 3.4 — Forces in the same direction add together; forces in opposite directions subtract.

Balanced vs. Unbalanced Forces

★ Balanced Forces

Equal forces acting in **opposite directions**. The net force is zero. The object doesn't change motion — if it's still, it stays still. If it's moving, it keeps moving the same.

Examples: Tug-of-war when nobody wins; a book sitting on a desk; you standing still on the floor.

★ Unbalanced Forces

Forces that are **not equal**. There's a net force. The object's motion **changes** — it speeds up, slows down, or changes direction.

Examples: Kicking a soccer ball; a car accelerating; a falling object.

✗ Common Mix-Up

"No force = no motion" is **wrong**. An object moving at constant velocity has *zero net force* on it. A hockey puck sliding on perfect ice keeps going forever without any forward force pushing it. Forces only change motion — they're not needed to maintain it.

? Quick Check 3.6

- Two friends push a stalled car. One pushes east with 200 N. The other pushes east with 150 N. What's the net force?
- If you push a desk with 50 N and friction pushes back with 50 N, will the desk move? Why?
- What's the difference between balanced and unbalanced forces?

3.7 Newton's First Law: Inertia

In the 1680s, Isaac Newton wrote three laws that became the foundation of all physics. The first one is sometimes called the **Law of Inertia**:

1 Newton's First Law

An object at rest stays at rest, and an object in motion stays in motion at the same speed and direction, unless acted on by an unbalanced force.

Inertia is the tendency of an object to **resist changes** in its motion. Objects don't want to start moving. And once they're moving, they don't want to stop. They just want to keep doing whatever they're already doing.

The more **mass** an object has, the more inertia it has. That's why it's easy to push a shopping cart but extremely hard to push a car. Both want to stay still — but the car wants it way more.

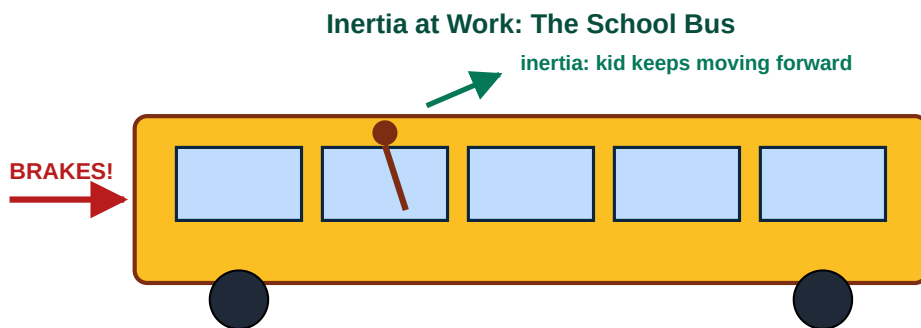


Figure 3.5 — When the bus brakes, the bus slows down — but YOU don't get the brake force directly. Your body keeps moving forward (inertia) until your seatbelt or the seat in front of you provides the force to stop you.

★ Inertia in Everyday Life

- The tablecloth trick — yank fast and the dishes stay put. They have too much inertia to move sideways with the cloth.
- Why seatbelts exist. The car stops in a crash, but YOU keep moving forward unless something stops you.
- Slamming on bike brakes and your phone flying out of your backpack — phone wants to keep going!
- Shaking ketchup down. The ketchup has inertia and lags behind when you slow your arm down quickly.

3.8 Newton's Second Law: $F = ma$

Newton's Second Law puts a number on what we just learned about inertia. It says how much force you need to make something accelerate.

2 Newton's Second Law

The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

$$\text{Force} = \text{Mass} \times \text{Acceleration}$$

$$F \text{ (N)} = m \text{ (kg)} \times a \text{ (m/s}^2\text{)}$$

This formula tells us two things:

→ More Force = More Acceleration

If you push the *same* object harder, it accelerates faster. Lightly tap a soccer ball and it rolls slowly. Crush it with a kick and it rockets across the field.

→ More Mass = Less Acceleration

If you use the *same* force on a heavier object, it accelerates less. Push a kid on a swing — easy. Push an adult on the same swing with the same force — much less movement.

Force is Proportional to Mass (for the same acceleration)

Here's another way to think about it: to make objects accelerate at the same rate, the heavier object needs more force. To push a shopping cart from 0 to 5 m/s might take 20 N. To push a car (way more mass) from 0 to 5 m/s in the same time would take thousands of newtons. **The more mass, the more force needed** — that's the connection to inertia.

! Did You Know?

NASA's Saturn V rocket that launched astronauts to the Moon had a mass of about **2.8 million kg** at liftoff. To get it accelerating off the launch pad, the engines produced about **35 million newtons** of thrust. Even with that much force, the rocket barely crawled at first — pure $F = ma$ in action.

? Quick Check 3.8

1. If a 5 kg object has a net force of 20 N on it, what is its acceleration?
2. If you double the mass of an object but keep the force the same, what happens to the acceleration?
3. Why does a baseball thrown by a pitcher accelerate faster than a bowling ball thrown with the same force?

3.9 Newton's Third Law: Action & Reaction

The last of Newton's laws is the most famous and the most misunderstood:

3 Newton's Third Law

For every action, there is an equal and opposite reaction.

Forces always come in **pairs**. If object A pushes on object B, then B is pushing back on A — with the exact same amount of force, in the opposite direction. Always. Every time.

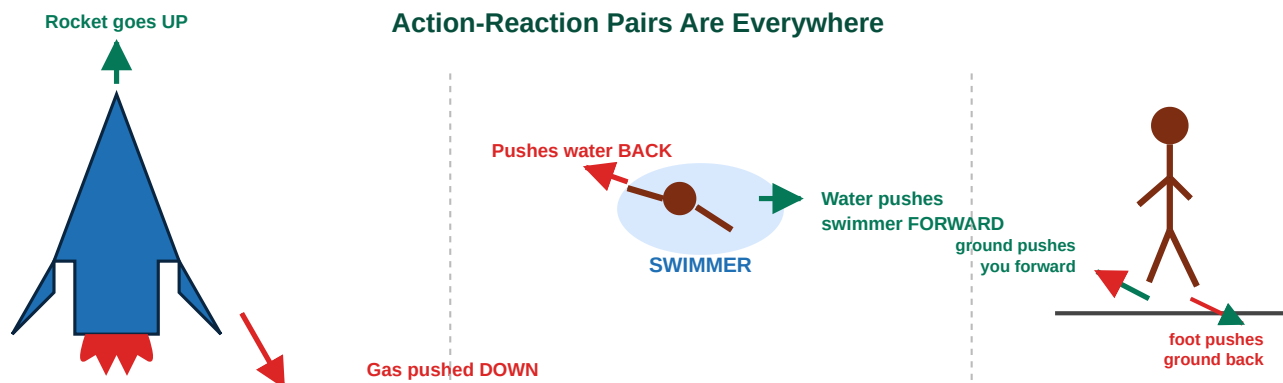


Figure 3.6 — Forces always come in pairs. A rocket pushes gas down → the gas pushes the rocket up. You push the ground backward → the ground pushes you forward.

Action	Reaction
You push the ground backward when you walk	The ground pushes you forward
A rocket pushes hot gas downward	The gas pushes the rocket upward
A swimmer pushes water backward	Water pushes the swimmer forward
A bat hits a baseball	The ball pushes back on the bat (you feel the impact)
Earth pulls you down with gravity	You pull Earth up with gravity (yes, really!)

✗ Common Mix-Up

If forces are always equal and opposite, why don't they cancel out? Because **they act on different objects!** When you push a wall, you exert force **ON** the wall and the wall exerts force **ON** you. These are not the same forces on the same object — they're on different objects. Forces only cancel when they act on the **SAME** object.

! Did You Know?

When you jump, you push down on Earth with your full weight — and Earth pushes you back up with the same force. Technically, Earth jumps up slightly when you jump down. But Earth has **5,972,000,000,000,000,000,000 kg of mass** (yes, really) so its acceleration is so small no instrument could ever measure it. $F = ma$ — when m is enormous, a is microscopic.

? Quick Check 3.9

1. When you push a wall with 30 N of force, how much force does the wall push back with?
2. If forces are always paired and equal, how can anything ever move? Explain.
3. Identify the action-reaction pair when a soccer player kicks a soccer ball.

★ CHAPTER 3 BIG IDEAS

- **Motion** is a change in position relative to a reference point. **Distance** is total path traveled; **displacement** is straight-line change in position.
- **Speed** = distance ÷ time. **Velocity** = speed with direction.
- **Acceleration** = change in velocity ÷ time. Includes speeding up, slowing down, OR changing direction.
- On a **distance-time graph**, slope = speed. On a **velocity-time graph**, slope = acceleration.
- A **force** is a push or pull, measured in **newtons (N)**. **Net force** is the total force after combining all forces on an object.
- **Balanced forces** (net = 0) → no change in motion. **Unbalanced forces** → motion changes.
- **Newton's First Law (Inertia)**: Objects don't change motion unless a force acts on them. More mass = more inertia.
- **Newton's Second Law**: $F = m \times a$. More force = more acceleration; more mass = less acceleration.
- **Newton's Third Law**: Every action force has an equal and opposite reaction force — acting on a different object.

I Can... (Self-Check)

- ☐ I can analyze and interpret data to identify patterns in the relationships between speed and distance, and velocity and acceleration. (*S8P3.a*)
- ☐ I can use Newton's Laws of Motion to explain the effects of balanced and unbalanced forces on the motion of an object. (*S8P3.b*)
- ☐ I can use evidence to argue that the amount of force needed to accelerate an object is proportional to its mass (inertia). (*S8P3.c*)